

PRD: Unified Smart Home & Energy Experience

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Overview & problem statement

We will build **HomeOne** (working name): one app, one control layer and one intelligence layer that make lighting, shades, controls, HVAC and energy systems behave as a single, energy-aware home.

The problem. The industry has connected individual products, but customers still experience them as separate solutions. Each category typically ships its own app, setup flow, schedules and notifications, so the home is "smart" device by device but not as a whole.

What this costs customers today:

- **Fragmentation:** multiple apps, logins and controls for one home; routines must be rebuilt in each.
- **Products that ignore each other:** shades don't help the HVAC, the HVAC doesn't know the tariff, lights don't know the room is empty.
- **Invisible energy:** customers can't see what each system costs, or act on it in the moment.
- **Complexity ceiling:** advanced automation needs installer or hobbyist skills, so most value stays unused.

The opportunity. Coordinating products around shared context — occupancy, time, weather, comfort and energy price — unlocks value no single product delivers alone: lower bills, better comfort, less effort, and a stickier customer relationship for the brand.

Problem statement: How can we bring multiple smart-home and energy capabilities together into a unified experience that improves convenience, energy efficiency and customer value?

Goals, non-goals and success metrics

Success means customers run their whole home from one place, save measurable energy without thinking about it, and stay with the ecosystem longer.

Goals

1. **Convenience:** one app and one set of scenes, schedules and voice commands across all product categories.
2. **Energy efficiency:** coordinate devices automatically around occupancy, weather and tariffs to cut energy use and cost.
3. **Customer value:** make savings and comfort visible, and create attach and recurring-revenue opportunities.
4. **Simplicity:** setup and automation a non-technical homeowner can complete without an installer.

Non-goals (v1)

- Building new hardware; v1 orchestrates existing products.
- Full security/camera platform (integrate later via partners).
- Direct utility billing or energy trading.

Success metrics (targets are proposals to validate with baseline data)

Metric	Definition	Target (12 months post-launch)
Unified-app adoption	% of active households using HomeOne as primary app	≥ 60%
Cross-category automations	Avg. active scenes/routines spanning ≥ 2 categories per home	≥ 3
Energy savings	Measured kWh reduction vs. household baseline	10–15%
Peak-load shift	% of eligible load moved out of peak tariff windows	≥ 20%
Setup time	Median time to onboard first 5 devices	< 15 min
Satisfaction	App NPS / CSAT	NPS ≥ 40
Support load	Support tickets per 1,000 homes	– 30%
Revenue	Attach rate of additional categories; premium plan conversion	+20% attach; 8% premium

Target users and personas

The primary user is the homeowner who wants comfort and savings without managing devices; installers and household members are secondary users the experience must also serve.

Persona	Who they are	Needs	Pain today
Comfort-first homeowner (primary)	Owens 2-4 smart categories, not technical	Home that "just works"; one place to control everything	Too many apps; routines break; unclear value
Energy-conscious owner	Has or plans solar, battery, EV or heat pump	See and cut energy cost; use own solar first	Energy data is siloed; can't act on tariffs
Household members & guests	Family, elderly parents, kids, visitors	Simple wall/voice control; limited access	Confused by apps; need permissions
Installer / integrator	Pro who designs and commissions systems	Fast commissioning, remote support, upsell	Separate tools per brand; costly truck rolls
Property manager / builder	Multi-unit or new-build developer	Standard package, fleet visibility	No portfolio view; inconsistent setups

Key use cases / user journeys

Each journey below needs at least two product categories working together — that coordination is the value no single product delivers.

1. **Good morning:** at wake time, shades open gradually, lights ramp to warm white, HVAC reaches comfort temperature 20 minutes before, and the app shows today's energy outlook.
2. **Leaving home:** geofence or one tap sets "Away" — lights off, HVAC to eco setback, shades close on the sunny side, and the user gets a confirmation with estimated savings.
3. **Hot afternoon (passive cooling):** forecast shows 35°C; shades close on south/west façades before peak sun so HVAC runs less.

4. **Peak-tariff response:** before a peak window the system pre-cools/pre-heats, shifts EV charging and appliance runs, and dims non-essential loads; the user can override any action.
5. **Solar self-consumption:** surplus solar is directed to battery, EV, water heater or pre-conditioning instead of exporting at low feed-in rates.
6. **Movie night / scenes:** one voice command or keypad press dims lights, lowers shades and adjusts temperature in the living room.
7. **Unoccupied room:** occupancy sensors turn off lights and relax HVAC set-points in empty rooms.
8. **Monthly energy insight:** a report shows cost by category, savings achieved by automations, and 2–3 personalised recommendations.
9. **Installer handover:** the installer commissions all products in one tool, applies a recommended scene package, and hands over the home to the owner's account.
10. **Guest / elderly access:** restricted profile with simple controls, plus alerts for unusual inactivity (opt-in).

Product scope & functional requirements

V1 delivers seven capability areas; P0 items are required for launch, P1 for the first major release after, P2 later.

ID	Area	Requirement	Priority
FR-1	Unified app	Single app and account to discover, control and monitor lighting, shades, keypads/controls, HVAC and energy devices	P0
FR-2	Unified app	Home → room → device model; any device appears in its room regardless of brand/protocol	P0
FR-3	Unified app	Home dashboard: comfort status, active scenes, live energy flow (grid, solar, battery, consumption)	P0
FR-4	Onboarding	Guided setup with QR/auto-discovery; import existing devices from brand apps/cloud accounts	P0

ID	Area	Requirement	Priority
FR-5	Scenes & automation	Cross-category scenes (e.g. "Movie", "Away") triggered by app, keypad, voice, schedule, geofence or sensor	P0
FR-6	Scenes & automation	Templates for common routines; no-code rule builder (if / when / then)	P0
FR-7	Scenes & automation	Recommended automations generated from usage patterns, accepted with one tap	P1
FR-8	Energy intelligence	Per-category and per-device energy consumption and cost (measured or estimated)	P0
FR-9	Energy intelligence	Time-of-use tariff input and automatic peak avoidance (pre-conditioning, load shifting)	P0
FR-10	Energy intelligence	Weather-aware shade/HVAC coordination for passive heating and cooling	P0
FR-11	Energy intelligence	Solar/battery/EV optimisation for self-consumption	P1
FR-12	Energy intelligence	Utility demand-response program enrolment with opt-out per event	P2
FR-13	Comfort	Comfort profiles per room and per person; system balances comfort vs. savings by a user-set slider	P1
FR-14	Controls	Physical keypads and wall controls run the same scenes as the app; work locally if internet is down	P0
FR-15	Voice & ecosystems	Works with major voice assistants and Matter controllers	P0
FR-16	Insights	Monthly energy & savings report with personalised recommendations	P1
FR-17	Notifications	One consolidated notification centre with priority levels and quiet hours	P0
FR-18	Users & access	Household members, guest and installer roles with granular permissions	P0

ID	Area	Requirement	Priority
FR-19	Installer tools	Commissioning, remote diagnostics and handover in one portal	P1
FR-20	Device health	Alerts for offline devices, low batteries, filter changes, firmware updates	P1
FR-21	Monetisation	Premium tier (advanced energy optimisation, extended history, AI assistant)	P2
FR-22	AI assistant	Natural-language control and questions ("Why was my bill higher?")	P2

Non-functional requirements

The home must keep working locally when the cloud doesn't, and every automation must be safe, fast and overridable.

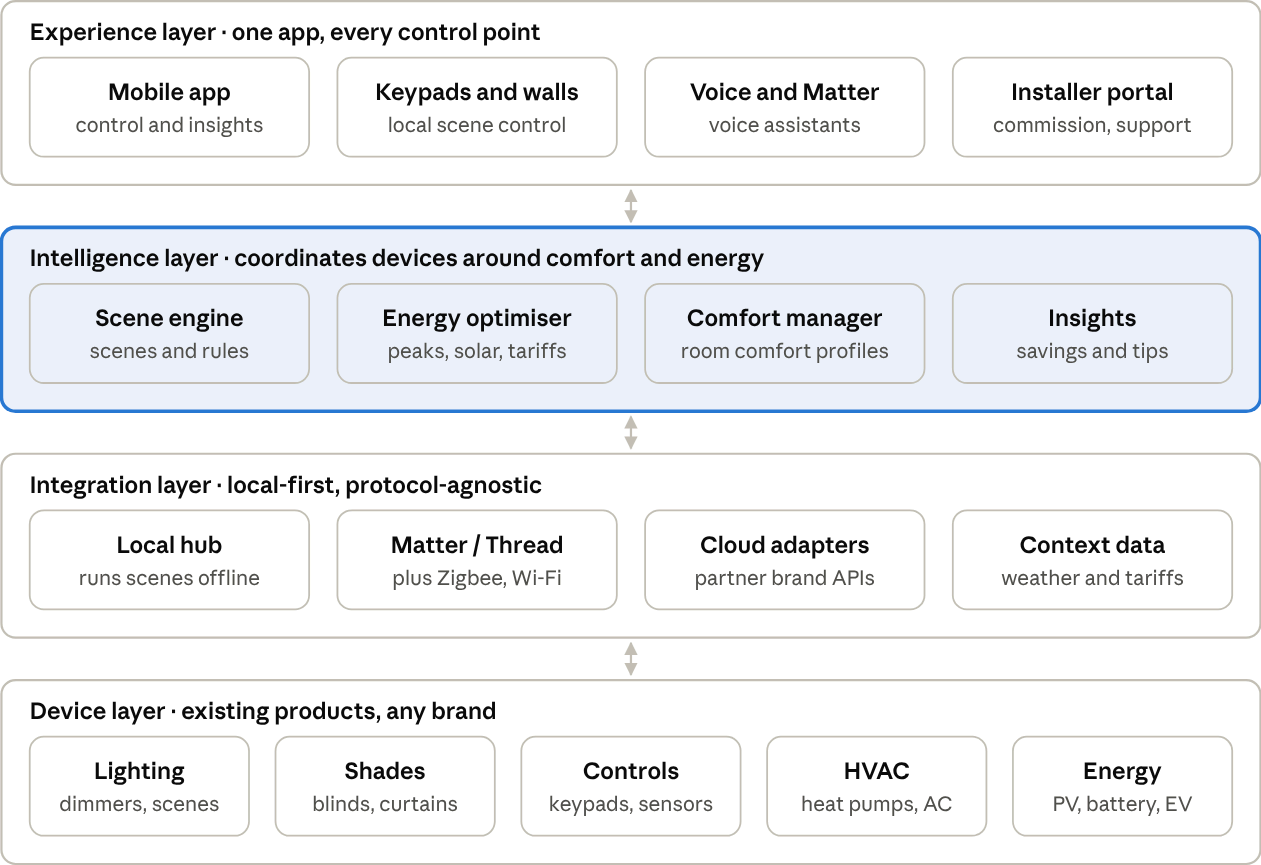
Area	Requirement
Reliability	Core scenes, keypads and schedules run locally on the hub during internet outages; 99.9% cloud uptime
Latency	Local control < 300 ms from press to action; app control < 1 s
Interoperability	Matter/Thread, Zigbee, Wi-Fi and vendor cloud APIs via an adapter layer; open API for partners
Security	Encrypted device and cloud traffic, MFA, signed OTA firmware, regular penetration testing
Privacy	Data minimisation; occupancy and energy data used only with explicit consent; export and delete on request; GDPR/regional compliance
Safety & control	Every automated action is visible, explainable and overridable; HVAC set-point limits never breached
Scalability	Support 200+ devices per home and millions of homes

Area	Requirement
Accessibility	WCAG 2.2 AA in the app; large-text and simple-mode options
Energy accuracy	Measured data where meters exist; estimates clearly labelled with $\pm$ range

Solution architecture overview

HomeOne adds a shared intelligence layer between the customer's control points and existing devices, so no product has to be replaced.

One intelligence layer coordinates every device category



HomeOne architecture · 4 layers

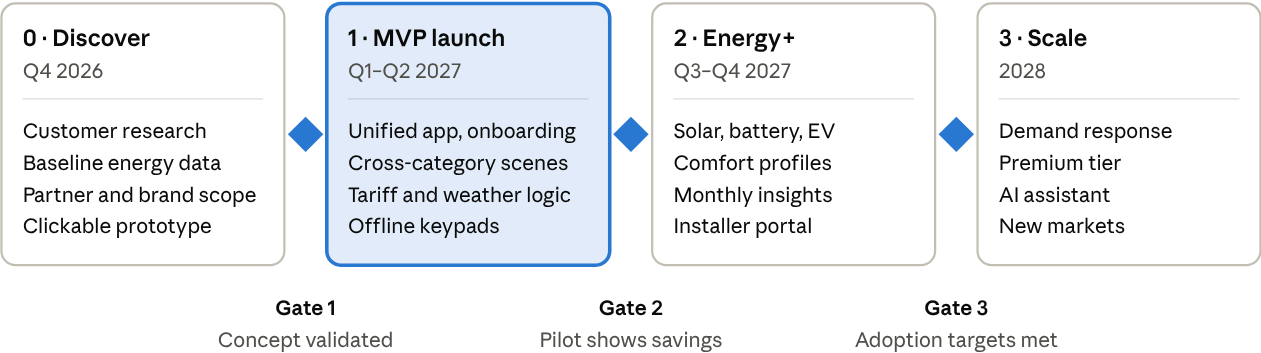
The intelligence layer (highlighted) is the differentiator: it sees occupancy, weather, tariffs and comfort together and decides what each device should do. Safety-critical scenes and keypads run on the local hub, so the home still works offline.

Release plan / roadmap

We ship the unified app and cross-category scenes first, then deepen energy optimisation once a pilot proves real savings.

Four phases take HomeOne from research to scale

Proposed timing for planning; phases not drawn to scale. Diamonds are go/no-go gates.



Release roadmap · 4 phases, 3 gates (proposed)

The MVP (highlighted) covers all P0 requirements; each gate is a go/no-go review against the success metrics above. Timing is a planning proposal to confirm with engineering.

Risks, dependencies and open questions

The biggest risk is trust: if automations feel unpredictable or savings can't be shown, customers will switch them off.

Risk	Impact	Mitigation
Third-party device integration is inconsistent	Broken experience, support cost	Certified-device list; Matter-first; adapter test suite
Automation feels "out of control"	Users disable features	Explain every action, easy override, comfort-vs-savings slider
Savings can't be proven	Weak value story	Baseline measurement at onboarding; transparent savings method
Privacy concerns about occupancy/energy data	Adoption, regulatory exposure	Consent-first design, local processing where possible

Risk	Impact	Mitigation
Tariff and utility data varies by region	Energy features limited	Start with 1-2 launch markets; manual tariff entry fallback
Migration from existing brand apps	Churn during transition	Import tool; run legacy apps in parallel for 6-12 months

Dependencies: hub/bridge hardware availability; partner APIs (HVAC, inverter, EV charger, utilities); voice-assistant certification; weather and tariff data providers.

Open questions

- ☐ Which launch market(s) and which tariff structures do we support first?
- ☐ Is a local hub mandatory, or is cloud-only acceptable for some product lines?
- ☐ Which third-party brands are in scope for v1 integration?
- ☐ Pricing model: free core + premium tier, hardware bundle, or installer subscription?
- ☐ Who owns the customer relationship when an installer commissions the home?
- ☐ What baseline method will we use to calculate and claim energy savings?